LECTURE - 13

13/3/26 (online)
Friday (continue prev)

Random variable & Prob. Distribution:

- Random variable that takes it outcomes from random experiment.
- Eg: we roll a dice and outcomes are: 1, 2, 3, 4, 5, 6;
Then 'x' will be a random variable where "X = number on dice".
- Since we don't know value of X (i.e: number about to be rolled)
∴ its called random variable.

* Random variable always has prob. distribution.

⇒ Simple prob vs Prob. Distribution:

Simple prob	Prob. Distribution
Answers one specific Question about any event.	This describes all possible outcomes & their probabilities together.
eg: Bag has red & blue balls. Simple prob would be to find "prob. of 2 red 1 blue ball".	eg. Bag has red & blue balls. X = no. of yellow balls drawn; then a table with X = 1, 2, 3, 4 ... with their prob. will be made.

⇒ Random variable has two types:

① Quantitative:

→ Discrete:
- Values are countable & finite
- f(x) is called "Probability distribution function" (PDF)
- $f(x) \geq 0$
- $\text{Sum}(f(x)) = 1$

→ Continuous:
- Values are in b/w an interval a & b (where $a, b \to -\infty \to \infty$)
- f(x) is called "Probability density function"
- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x) \cdot dx = 1$

② Qualitative

⟹ Discrete Random Variable (Examples):

Q: Find prob. distribution of the sum of two dots when a fair die is thrown:

$$S = \begin{bmatrix} (1,1), (1,2) (1,3) \ldots\ldots (1,6) \\ (2,1), (2,2) \text{———} (2,6) \\ (3,1) \ldots\ldots\ldots (3,6) \\ \vdots \\ (6,1) \ldots\ldots\ldots (6,6) \end{bmatrix}$$

Sum of two dots = {2, 3, 4, 5, 6, 7, 3, 4, 5, 6, 7, 8, 4, 5, 6, 7, 8, 9, 5, 6, 7, 8, 9, 10, 6, 7, 8, 9, 10, 11, 7, 8, 9, 10, 11, 12}

Now we'll find $f(x)$ or $P(X = x)$:

Tabular Form: X = sum of dots

X	Tally	freq.	f(x)
2	|	1	1/36
3	||	2	2/36
4	|||	3	3/36
5	||||	4	4/36
6	~~||||~~	5	5/36
7	~~||||~~ |	6	6/36
8	~~||||~~	5	5/36
9	||||	4	4/36
10	|||	3	3/36
11	||	2	2/36
12	|	1	1/36
		$\Sigma = 36$	$\Sigma = 1$

Q: Suppose X has a prob. distribution given by:

X	-1	0	1
f(x)	3c	3c	6c

i) Determine 'c'

* For f(X) to be a prob. distribution func., $\text{Sum}(f(x)) = 1$.

Ans: $3c + 3c + 6c = 1$

$12c = 1$

$\boxed{c = 1/12}$

(ii) Prob. distribution of $y = 2x + 1$?

By putting value of 'c' in dues. table

Ans:

X	-1	0	1
f(x)	3/12	3/12	6/12

* put values of 'x' in $y = 2x + 1$ for new table.

∴

y	-1	1	3
f(y)	3/12	3/12	6/12

random variable

e.g: $x = 1$, $y = 2(1) + 1$

$y = \boxed{3}$

→ This remains same as 'X' table (as this is prob. distr.)

Q: Determine prob. distribution of a random variable X, where X is no. of red Balls. A bag contains 4 red & 6 black balls. A sample of 4 balls is selected from bag w/out replacement.

Ans: X = no. of red balls ∴ X = 0, 1, 2, 3, 4. and their probs. are:

$$f(0) \Rightarrow P(X=0) \longrightarrow \frac{({}^4C_0)({}^6C_4)}{{}^{10}C_4} = \frac{15}{210} = 0.0714$$

$$f(1) = P(X=1) \longrightarrow \frac{({}^4C_1)({}^6C_3)}{{}^{10}C_4} = 0.3809$$

$$f(2) = P(X=2) \rightarrow \frac{({}^4C_2)({}^6C_2)}{{}^{10}C_4} = 0.4286$$

$$f(3) = P(X=3) \rightarrow \frac{({}^4C_3)({}^6C_1)}{{}^{10}C_4} = 0.1142$$

$f(4) = P(X=4) \rightarrow \frac{({}^4C_4)({}^6C_0)}{{}^{10}C_4} = 0.0047$

In tabular form:

X	0	1	2	3	4
f(x)	0.0714	0.3809	0.4286	0.1142	0.0047

Q: Same type question on page# 10 of "Random variable & its prob. distribution" PDF in GCR.

Q: Determine value 'c' so each of the following func. can serve as a prob. distri. of the discrete random variable X:

(a) $f(x) = c(x^2+4)$ for $x = 0, 1, 2, 3$.

* Putting values of x in $f(x)$.

$f(0) = c(0^2+4) = 4c$

$f(1) = c(1^2+4) = 5c$

$f(2) = c(2^2+4) = 8c$

$f(3) = c(3^2+4) = 13c$

* since the $f(x)$ is prob. distri. func. ∴ sum of $f(x)$ should be $= 1$.

$f(0) + f(1) + f(2) + f(3) \rightarrow 4c + 5c + 8c + 13c = 1$

$30c = 1$

$\boxed{c = 1/30}$

(b) $f(x) = c\,({}^2C_x)({}^3C_{3-x})$, $x = 0, 1, 2 \cdots$

* put values of x in $f(x)$

$f(0) = c({}^2C_0)({}^3C_{3-0}) = c(1)(1) = c$

$f(1) = c({}^2C_1)({}^3C_{3-1}) = c(2)(3) = 6c$

$f(2) = 3c$

Since prob. distr. func ∴

$c + 3c + 6c = 1$

$10c = 1$

$\boxed{c = 1/10}$

⇒ Continuous Random variable Questions:

Q: In a life of Battery follow the follow probability density func. as $f(x) = 1/4\, e^{-x/4}$, $0 < x < \infty$.

* Integrating $f(x)$ to see if it's $= 1$ to be a prob. density func.

$= \int_0^\infty \frac{1}{4} e^{-x/4}\, dx$ — $e^{-x/4}$ as it is and divided by its derivative

$= \frac{1}{4}\left[\frac{e^{-x/4}}{-1/4}\right]_0^\infty \rightarrow \frac{1}{\cancel{4}}(\cancel{-4})(e^{-\infty} - e^0)$

$= -(0-1) = \boxed{+1}$

$f(x) = 1$ ✓ ∴ $f(x)$ is prob. density func.

(b) Find the prob. that battery having life less than 7.

Ans: $P(x<7)$ ∴ limits $0 \leftrightarrow 7$

$\int_0^7 f(x) \cdot dx = \int_0^7 \frac{1}{4} e^{-x/4} \cdot dx$.

* After integrating & limits → answer $= \boxed{0.8262}$

Q: Find value of 'k' for which $f(x) = ke^{-2.5x}$ is a prob. density func., $0 < x < \infty$ where $k > 0$.

(a) Find the Prob. when $P(x < 5)$.

$\int_0^5 ke^{2.5x} \cdot dx \rightarrow k\left[\frac{e^{-2.5x}}{-2.5}\right]_0^5 \rightarrow \frac{k}{-2.5}\left[e^{-2.5(5)} - e^{(-2.5)(0)}\right]$

$= \frac{k(0.9999)}{+2.5} \rightarrow (1)$

* For value of k $f(x) = 1$

$1 = k\int_0^\infty e^{-2.5x} \cdot dx \rightarrow 1 = k\left[\frac{e^{-2.5x}}{-2.5}\right]_0^\infty \xrightarrow{\text{after Integration}} 0.4k = 1$

$\boxed{k = 2.5}$

* putting $k = 2.5$ in ① for prob. when $x < 5$.

$= \frac{\cancel{+2.5}}{\cancel{+2.5}} (0.999)$ answer = $\boxed{0.999}$

(b) Prob. when $x > 3$.

* Since $x > 3$ in this case $\therefore$ limits would be $3 \to \infty$.

and $k = 2.5$: answer after integrating this $\boxed{2.5 \int_3^\infty e^{-2.5x} \, dx}$

(c) Find prob. when $2 < x < 6$)

Integrate. $k \int_2^6 e^{-2.5x} \, dx$

NOTE:

$P(x \leq 7) = P(x < 7) + P(X = 7)$

$= P(x < 7) + 0$

$P(x \leq 7) = P(x < 7)$

always zero since in continuous ; x can't be equal to smth therefore less than is same as less than equal to.